$$\alpha_1 = \sqrt{\frac{ma^2E_1}{2\hbar^2}} \simeq 1.9 \quad \Longrightarrow \quad E_1 \simeq \frac{7.22\hbar^2}{ma^2}. \tag{4.113}$$

4.8 The Harmonic Oscillator

The harmonic oscillator is one of those few problems that are important to all branches of physics. It provides a useful model for a variety of vibrational phenomena that are encountered, for instance, in classical mechanics, electrodynamics, statistical mechanics, solid state, atomic, nuclear, and particle physics. In quantum mechanics, it serves as an invaluable tool to illustrate the basic concepts and the formalism.

The Hamiltonian of a particle of mass m which oscillates with an angular frequency ω under the influence of a one-dimensional harmonic potential is

$$\hat{H} = \frac{\hat{P}^2}{2m} + \frac{1}{2}m\omega^2\hat{X}^2. \tag{4.114}$$

The problem is how to find the energy eigenvalues and eigenstates of this Hamiltonian. This problem can be studied by means of two separate methods. The first method, called the *analytic method*, consists in solving the time-independent Schrödinger equation (TISE) for the Hamiltonian (4.114). The second method, called the *ladder* or *algebraic method*, does not deal with solving the Schrödinger equation, but deals instead with operator algebra involving operators known as the *creation* and *annihilation* or *ladder* operators; this method is in essence a matrix formulation, because it expresses the various quantities in terms of matrices. In our presentation, we are going to adopt the second method, for it is more straightforward, more elegant and much simpler than solving the Schrödinger equation. Unlike the examples seen up to now, solving the Schrödinger equation for the potential $V(x) = \frac{1}{2}m\omega x^2$ is no easy job. Before embarking on the second method, let us highlight the main steps involved in the first method.

Brief outline of the analytic method

This approach consists in using the power series method to solve the following differential (Schrödinger) equation:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + \frac{1}{2}m\omega^2x^2\psi(x) = E\psi(x), \tag{4.115}$$

which can be reduced to

$$\frac{d^2\psi(x)}{dx^2} + \left(\frac{2mE}{\hbar^2} - \frac{x^2}{x_0^4}\right)\psi(x) = 0, \tag{4.116}$$

where $x_0 = \sqrt{\hbar/(m\omega)}$ is a constant that has the dimensions of length; it sets the length scale of the oscillator, as will be seen later. The solutions of differential equations like (4.116) have been worked out by our mathematician colleagues well before the arrival of quantum mechanics (the solutions are expressed in terms of some special functions, the Hermite polynomials). The occurrence of the term $x^2\psi(x)$ in (4.116) suggests trying a Gaussian type solution[3]: $\psi(x) =$

[3] Solutions of the type $\psi(x) = f(x)\exp(x^2/2x_0^2)$ are physically unacceptable, for they diverge when $x \longrightarrow \pm\infty$.

$f(x)\exp(-x^2/2x_0^2)$, where $f(x)$ is a function of x. Inserting this trial function into (4.116), we obtain a differential equation for $f(x)$. This new differential equation can be solved by expanding $f(x)$ out in a power series (i.e., $f(x) = \sum_{n=0}^{\infty} a_n x^n$, where a_n is just a coefficient), which when inserted into the differential equation leads to a recursion relation. By demanding the power series of $f(x)$ to terminate at some finite value of n (because the wave function $\psi(x)$ has to be finite everywhere, notably when $x \longrightarrow \pm\infty$), the recursion relation yields an expression for the energy eigenvalues which are discrete or quantized:

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega \qquad (n = 0, 1, 2, 3, \ \ldots). \tag{4.117}$$

After some calculations, we can show that the wave functions that are physically acceptable and that satisfy (4.116) are given by

$$\psi_n(x) = \frac{1}{\sqrt{\sqrt{\pi}2^n n! x_0}} e^{-x^2/2x_0^2} H_n\left(\frac{x}{x_0}\right), \tag{4.118}$$

where $H_n(y)$ are nth order polynomials called *Hermite polynomials*:

$$H_n(y) = (-1)^n e^{y^2} \frac{d^n}{dy^n} e^{-y^2}. \tag{4.119}$$

From this relation it is easy to calculate the first few polynomials:

$$\begin{aligned} &H_0(y) = 1, && H_1(y) = 2y, \\ &H_2(y) = 4y^2 - 2, && H_3(y) = 8y^3 - 12y, \\ &H_4(y) = 16y^4 - 48y^2 + 12, && H_5(y) = 32y^5 - 160y^3 + 120y. \end{aligned} \tag{4.120}$$

We will deal with the physical interpretations of the harmonic oscillator results when we study the second method.

Algebraic method

Let us now show how to solve the harmonic oscillator eigenvalue problem using the algebraic method. For this, we need to rewrite the Hamiltonian (4.114) in terms of the two Hermitian, dimensionless operators $\hat{p} = \hat{P}/\sqrt{m\hbar\omega}$ and $\hat{q} = \hat{X}\sqrt{m\omega/\hbar}$:

$$\hat{H} = \frac{\hbar\omega}{2}(\hat{p}^2 + \hat{q}^2), \tag{4.121}$$

and then introduce two non-Hermitian, dimensionless operators:

$$\hat{a} = \frac{1}{\sqrt{2}}(\hat{q} + i\,\hat{p}), \qquad \hat{a}^\dagger = \frac{1}{\sqrt{2}}(\hat{q} - i\,\hat{p}). \tag{4.122}$$

The physical meaning of the operators $\hat{a}$ and $\hat{a}^\dagger$ will be examined later. Note that

$$\hat{a}^\dagger\hat{a} = \frac{1}{2}(\hat{q} - i\,\hat{p})(\hat{q} + i\,\hat{p}) = \frac{1}{2}(\hat{q}^2 + \hat{p}^2 + i\hat{q}\,\hat{p} - i\,\hat{p}\hat{q}) = \frac{1}{2}(\hat{q}^2 + \hat{p}^2) + \frac{i}{2}[\hat{q},\ \hat{p}], \tag{4.123}$$

where, using $[\hat{X}, \hat{P}] = i\hbar$, we can verify that the commutator between $\hat{q}$ and $\hat{p}$ is

$$[\hat{q},\ \hat{p}] = \left[\sqrt{\frac{m\omega}{\hbar}}\hat{X}, \frac{1}{\sqrt{\hbar m\omega}}\hat{P}\right] = \frac{1}{\hbar}\left[\hat{X}, \hat{P}\right] = i; \tag{4.124}$$

hence

$$\hat{a}^\dagger \hat{a} = \frac{1}{2}(\hat{q}^2 + \hat{p}^2) - \frac{1}{2} \tag{4.125}$$

or

$$\frac{1}{2}(\hat{q}^2 + \hat{p}^2) = \hat{a}^\dagger \hat{a} + \frac{1}{2}. \tag{4.126}$$

Inserting (4.126) into (4.121) we obtain

$$\boxed{\hat{H} = \hbar\omega\left(\hat{a}^\dagger \hat{a} + \frac{1}{2}\right) = \hbar\omega\left(\hat{N} + \frac{1}{2}\right) \qquad \text{with} \quad \hat{N} = \hat{a}^\dagger \hat{a},} \tag{4.127}$$

where $\hat{N}$ is known as the *number* operator or *occupation number* operator, which is clearly Hermitian.

Let us now derive the commutator $[\hat{a}, \hat{a}^\dagger]$. Since $[\hat{X}, \hat{P}] = i\hbar$ we have $[\hat{q}, \hat{p}] = \frac{1}{\hbar}[\hat{X}, \hat{P}] = i$; hence

$$[\hat{a}, \hat{a}^\dagger] = \frac{1}{2}\left[\hat{q} + i\hat{p}, \hat{q} - i\hat{p}\right] = -i\left[\hat{q}, \hat{p}\right] = 1 \tag{4.128}$$

or

$$\boxed{[\hat{a}, \hat{a}^\dagger] = 1.} \tag{4.129}$$

4.8.1 Energy Eigenvalues

Note that $\hat{H}$ as given by (4.127) commutes with $\hat{N}$, since $\hat{H}$ is linear in $\hat{N}$. Thus, $\hat{H}$ and $\hat{N}$ can have a set of joint eigenstates, to be denoted by $\mid n\rangle$:

$$\hat{N} \mid n\rangle = n \mid n\rangle \tag{4.130}$$

and

$$\hat{H} \mid n\rangle = E_n \mid n\rangle; \tag{4.131}$$

the states $\mid n\rangle$ are called energy eigenstates. Combining (4.127) and (4.131), we obtain the energy eigenvalues at once:

$$\boxed{E_n = \left(n + \frac{1}{2}\right)\hbar\omega.} \tag{4.132}$$

We will show later that n is a *positive integer*; it cannot have negative values.

The physical meaning of the operators $\hat{a}$, $\hat{a}^\dagger$, and $\hat{N}$ can now be clarified. First, we need the following two commutators that can be extracted from (4.129) and (4.127):

$$\boxed{[\hat{a}, \hat{H}] = \hbar\omega\hat{a}, \qquad [\hat{a}^\dagger, \hat{H}] = -\hbar\omega\hat{a}^\dagger.} \tag{4.133}$$

These commutation relations along with (4.131) lead to

$$\hat{H}\left(\hat{a} \mid n\rangle\right) \quad = (\hat{a}\hat{H} - \hbar\omega\hat{a}) \mid n\rangle \quad = (E_n - \hbar\omega)\left(\hat{a} \mid n\rangle\right), \tag{4.134}$$

$$\hat{H}\left(\hat{a}^\dagger \mid n\rangle\right) \quad = (\hat{a}^\dagger\hat{H} + \hbar\omega\hat{a}^\dagger) \mid n\rangle \; = (E_n + \hbar\omega)\left(\hat{a}^\dagger \mid n\rangle\right). \tag{4.135}$$

Thus, $\hat{a} \mid n\rangle$ and $\hat{a}^\dagger \mid n\rangle$ are eigenstates of $\hat{H}$ with eigenvalues $(E_n - \hbar\omega)$ and $(E_n + \hbar\omega)$, respectively. So the actions of $\hat{a}$ and $\hat{a}^\dagger$ on $\mid n\rangle$ generate new energy states that are lower and

higher by one unit of $\hbar\omega$, respectively. As a result, $\hat{a}$ and $\hat{a}^\dagger$ are respectively known as the *lowering* and *raising* operators, or the *annihilation* and *creation* operators; they are also known as the *ladder* operators.

Let us now find out how the operators $\hat{a}$ and $\hat{a}^\dagger$ act on the energy eigenstates $|\, n\rangle$. Since $\hat{a}$ and $\hat{a}^\dagger$ do not commute with $\hat{N}$, the states $|\, n\rangle$ are eigenstates neither to $\hat{a}$ nor to $\hat{a}^\dagger$. Using (4.129) along with $[\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}$, we can show that

$$\boxed{[\hat{N}, \hat{a}] = -\hat{a}, \qquad [\hat{N}, \hat{a}^\dagger] = \hat{a}^\dagger;} \tag{4.136}$$

hence $\hat{N}\hat{a} = \hat{a}(\hat{N} - 1)$ and $\hat{N}\hat{a}^\dagger = \hat{a}^\dagger(\hat{N} + 1)$. Combining these relations with (4.130), we obtain

$$\hat{N}\left(\hat{a} \mid n\rangle\right) = \hat{a}(\hat{N} - 1) \mid n\rangle = (n - 1)\left(\hat{a} \mid n\rangle\right), \tag{4.137}$$

$$\hat{N}\left(\hat{a}^\dagger \mid n\rangle\right) = \hat{a}^\dagger(\hat{N} + 1) \mid n\rangle = (n + 1)(\hat{a}^\dagger \mid n\rangle). \tag{4.138}$$

These relations reveal that $\hat{a} \mid n\rangle$ and $\hat{a}^\dagger \mid n\rangle$ are eigenstates of $\hat{N}$ with eigenvalues $(n - 1)$ and $(n + 1)$, respectively. This implies that when $\hat{a}$ and $\hat{a}^\dagger$ operate on $|\, n\rangle$, respectively, they decrease and increase n by one unit. That is, while the action of $\hat{a}$ on $|\, n\rangle$ generates a new state $|\, n - 1\rangle$ (i.e., $\hat{a} \mid n\rangle \sim |\, n - 1\rangle$), the action of $\hat{a}^\dagger$ on $|\, n\rangle$ generates $|\, n + 1\rangle$. Hence from (4.137) we can write

$$\hat{a} \mid n\rangle = c_n \mid n - 1\rangle, \tag{4.139}$$

where c_n is a constant to be determined from the requirement that the states $|\, n\rangle$ be normalized for all values of n. On the one hand, (4.139) yields

$$\left(\langle n \mid \hat{a}^\dagger\right) \cdot \left(\hat{a} \mid n\rangle\right) = \langle n \mid \hat{a}^\dagger\hat{a} \mid n\rangle = |c_n|^2 \langle n - 1 \mid n - 1\rangle = |c_n|^2 \tag{4.140}$$

and, on the other hand, (4.130) gives

$$\left(\langle n \mid \hat{a}^\dagger\right) \cdot \left(\hat{a} \mid n\rangle\right) = \langle n \mid \hat{a}^\dagger\hat{a} \mid n\rangle = n\langle n \mid n\rangle = n. \tag{4.141}$$

When combined, the last two equations yield

$$|c_n|^2 = n. \tag{4.142}$$

This implies that n, which is equal to the norm of $\hat{a} \mid n\rangle$ (see (4.141)), *cannot be negative*, $n \geq 0$, since the norm is a positive quantity. Substituting (4.142) into (4.139) we end up with

$$\boxed{\hat{a} \mid n\rangle = \sqrt{n} \mid n - 1\rangle.} \tag{4.143}$$

This equation shows that repeated applications of the operator $\hat{a}$ on $|\, n\rangle$ generate a sequence of eigenvectors $|\, n - 1\rangle$, $|\, n - 2\rangle$, $|\, n - 3\rangle$, ... Since $n \geq 0$ and since $\hat{a} \mid 0\rangle = 0$, this sequence has to terminate at $n = 0$; this is true if we start with an integer value of n. But if we start with a noninteger n, the sequence will not terminate; hence it leads to eigenvectors with negative values of n. But as shown above, since n cannot be negative, we conclude that n has to be a *nonnegative integer*.

Now, we can easily show, as we did for (4.143), that

$$\boxed{\hat{a}^\dagger \mid n\rangle = \sqrt{n+1} \mid n+1\rangle.} \tag{4.144}$$

This implies that repeated applications of $\hat{a}^\dagger$ on $\mid n\rangle$ generate an infinite sequence of eigenvectors $\mid n+1\rangle$, $\mid n+2\rangle$, $\mid n+3\rangle$, ... Since n is a positive integer, the energy spectrum of a harmonic oscillator as specified by (4.132) is therefore *discrete*:

$$\boxed{E_n = \left(n + \frac{1}{2}\right)\hbar\omega \qquad (n = 0, 1, 2, 3, \ldots).} \tag{4.145}$$

This expression is similar to the one obtained from the first method (see Eq. (4.117)). The energy spectrum of the harmonic oscillator consists of energy levels that are equally spaced: $E_{n+1} - E_n = \hbar\omega$. This is Planck's famous equidistant energy idea—the energy of the radiation emitted by the oscillating charges (from the inside walls of the cavity) must come only in bundles (quanta) that are integral multiples of $\hbar\omega$—which, as mentioned in Chapter 1, led to the birth of quantum mechanics.

As expected for bound states of one-dimensional potentials, the energy spectrum is both discrete and nondegenerate. Once again, as in the case of the infinite square well potential, we encounter the zero-point energy phenomenon: the lowest energy eigenvalue of the oscillator is not zero but is instead equal to $E_0 = \hbar\omega/2$. It is called the *zero-point energy* of the oscillator, for it corresponds to $n = 0$. The zero-point energy of bound state systems cannot be zero, otherwise it would violate the uncertainty principle. For the harmonic oscillator, for instance, the classical minimum energy corresponds to $x = 0$ and $p = 0$; there would be no oscillations in this case. This would imply that we know simultaneously and with absolute precision both the position and the momentum of the system. This would contradict the uncertainty principle.

4.8.2 Energy Eigenstates

The algebraic or operator method can also be used to determine the energy eigenvectors. First, using (4.144), we see that the various eigenvectors can be written in terms of the ground state $\mid 0\rangle$ as follows:

$$\mid 1\rangle = \hat{a}^\dagger \mid 0\rangle, \tag{4.146}$$

$$\mid 2\rangle = \frac{1}{\sqrt{2}}\hat{a}^\dagger \mid 1\rangle = \frac{1}{\sqrt{2!}}\left(\hat{a}^\dagger\right)^2 \mid 0\rangle, \tag{4.147}$$

$$\mid 3\rangle = \frac{1}{\sqrt{3}}\hat{a}^\dagger \mid 2\rangle = \frac{1}{\sqrt{3!}}\left(\hat{a}^\dagger\right)^3 \mid 0\rangle, \tag{4.148}$$

$$\vdots$$

$$\mid n\rangle = \frac{1}{\sqrt{n}}\hat{a}^\dagger \mid n-1\rangle = \frac{1}{\sqrt{n!}}\left(\hat{a}^\dagger\right)^n \mid 0\rangle. \tag{4.149}$$

So, to find any excited eigenstate $\mid n\rangle$, we need simply to operate $\hat{a}^\dagger$ on $\mid 0\rangle$ n successive times.

Note that any set of kets $\mid n\rangle$ and $\mid n'\rangle$, corresponding to different eigenvalues, must be orthogonal, $\langle n' \mid n\rangle \sim \delta_{n',n}$, since $\hat{H}$ is Hermitian and none of its eigenstates is degenerate.

Moreover, the states $\mid 0\rangle, \mid 1\rangle, \mid 2\rangle, \mid 3\rangle, \ldots, \mid n\rangle, \ldots$ are *simultaneous eigenstates* of $\hat{H}$ and $\hat{N}$; the set $\{\mid n\rangle\}$ constitutes an orthonormal and complete basis:

$$\langle n' \mid n\rangle = \delta_{n',n}, \qquad \sum_{n=0}^{+\infty} \mid n\rangle\langle n \mid = 1. \tag{4.150}$$

4.8.3 Energy Eigenstates in Position Space

Let us now determine the harmonic oscillator wave function in the position representation.

Equations (4.146) to (4.149) show that, knowing the ground state wave function, we can determine any other eigenstate by successive applications of the operator $a^\dagger$ on the ground state. So let us first determine the ground state wave function in the position representation.

The operator $\hat{p}$, defined by $\hat{p} = \hat{P}/\sqrt{m\hbar\omega}$, is given in the position space by

$$\hat{p} = -\frac{i\hbar}{\sqrt{m\hbar\omega}}\frac{d}{dx} = -ix_0\frac{d}{dx}, \tag{4.151}$$

where, as mentioned above, $x_0 = \sqrt{\hbar/(m\omega)}$ is a constant that has the dimensions of length; it sets the length scale of the oscillator. We can easily show that the annihilation and creation operators $\hat{a}$ and $\hat{a}^\dagger$, defined in (4.122), can be written in the position representation as

$$\hat{a} = \frac{1}{\sqrt{2}}\left(\frac{\hat{X}}{x_0} + x_0\frac{d}{dx}\right) = \frac{1}{\sqrt{2}x_0}\left(\hat{X} + x_0^2\frac{d}{dx}\right), \tag{4.152}$$

$$\hat{a}^\dagger = \frac{1}{\sqrt{2}}\left(\frac{\hat{X}}{x_0} - x_0\frac{d}{dx}\right) = \frac{1}{\sqrt{2}x_0}\left(\hat{X} - x_0^2\frac{d}{dx}\right). \tag{4.153}$$

Using (4.152) we can write the equation $\hat{a} \mid 0\rangle = 0$ in the position space as

$$\langle x|\hat{a} \mid 0\rangle = \frac{1}{\sqrt{2}x_0}\langle x|\hat{X} + x_0^2\frac{d}{dx} \mid 0\rangle = \frac{1}{\sqrt{2}x_0}\left(x\,\psi_0(x) + x_0^2\frac{d\psi_0(x)}{dx}\right) = 0; \tag{4.154}$$

hence

$$\frac{d\psi_0(x)}{dx} = -\frac{x}{x_0^2}\psi_0(x), \tag{4.155}$$

where $\psi_0(x) = \langle x \mid 0\rangle$ represents the ground state wave function. The solution of this differential equation is

$$\psi_0(x) = A\exp\left(-\frac{x^2}{2x_0^2}\right), \tag{4.156}$$

where A is a constant that can be determined from the normalization condition

$$1 = \int_{-\infty}^{+\infty} dx\,|\psi_0(x)|^2 = A^2\int_{-\infty}^{+\infty} dx\exp\left(-\frac{x^2}{x_0^2}\right) = A^2\sqrt{\pi}x_0; \tag{4.157}$$

hence $A = (m\omega/(\pi\hbar))^{1/4} = 1/\sqrt{\sqrt{\pi}x_0}$. The normalized ground state wave function is then given by

$$\boxed{\psi_0(x) = \frac{1}{\sqrt{\sqrt{\pi}x_0}} \exp\left(-\frac{x^2}{2x_0^2}\right).} \tag{4.158}$$

This is a Gaussian function.

We can then obtain the wave function of any excited state by a series of applications of $\hat{a}^\dagger$ on the ground state. For instance, the first excited state is obtained by one single application of the operator $\hat{a}^\dagger$ of (4.153) on the ground state:

$$\begin{aligned} \langle x \mid 1\rangle &= \langle x|\hat{a}^\dagger \mid 0\rangle = \frac{1}{\sqrt{2}x_0}\left(x - x_0^2\frac{d}{dx}\right)\langle x \mid 0\rangle \\ &= \frac{1}{\sqrt{2}x_0}\left(x - x_0^2\left(-\frac{x}{x_0^2}\right)\right)\psi_0(x) = \frac{\sqrt{2}}{x_0}x\,\psi_0(x) \end{aligned} \tag{4.159}$$

or

$$\psi_1(x) = \frac{\sqrt{2}}{x_0}x\,\psi_0(x) = \sqrt{\frac{2}{\sqrt{\pi}x_0^3}}x\exp\left(-\frac{x^2}{2x_0^2}\right). \tag{4.160}$$

As for the eigenstates of the second and third excited states, we can obtain them by applying $\hat{a}^\dagger$ on the ground state twice and three times, respectively:

$$\langle x \mid 2\rangle = \frac{1}{\sqrt{2!}}\langle x|\left(a^\dagger\right)^2 \mid 0\rangle = \frac{1}{\sqrt{2!}}\left(\frac{1}{\sqrt{2}x_0}\right)^2\left(x - x_0^2\frac{d}{dx}\right)^2\psi_0(x), \tag{4.161}$$

$$\langle x \mid 3\rangle = \frac{1}{\sqrt{3!}}\langle x|\left(a^\dagger\right)^3 \mid 0\rangle = \frac{1}{\sqrt{3!}}\left(\frac{1}{\sqrt{2}x_0}\right)^3\left(x - x_0^2\frac{d}{dx}\right)^3\psi_0(x) \tag{4.162}$$

or

$$\psi_2(x) = \frac{1}{\sqrt{2\sqrt{\pi}x_0}}\left(\frac{2x^2}{x_0^2} - 1\right)\exp\left(-\frac{x^2}{2x_0^2}\right), \quad \psi_3(x) = \frac{1}{\sqrt{3\sqrt{\pi}x_0}}\left(\frac{2x^3}{x_0^3} - \frac{3x}{x_0}\right)\exp\left(-\frac{x^2}{2x_0^2}\right). \tag{4.163}$$

Similarly, using (4.149), (4.153), and (4.158), we can easily infer the energy eigenstate for the nth excited state:

$$\langle x \mid n\rangle = \frac{1}{\sqrt{n!}}\langle x|\left(a^\dagger\right)^n \mid 0\rangle = \frac{1}{\sqrt{n!}}\left(\frac{1}{\sqrt{2}x_0}\right)^n\left(x - x_0^2\frac{d}{dx}\right)^n\psi_0(x), \tag{4.164}$$

which in turn can be rewritten as

$$\boxed{\psi_n(x) = \frac{1}{\sqrt{\sqrt{\pi}2^n n!}}\frac{1}{x_0^{n+1/2}}\left(x - x_0^2\frac{d}{dx}\right)^n\exp\left(-\frac{x^2}{2x_0^2}\right).} \tag{4.165}$$

In summary, by successive applications of $\hat{a}^\dagger = (\hat{X} - x_0^2 d/dx)/(\sqrt{2}x_0)$ on $\psi_0(x)$, we can find the wave function of any excited state $\psi_n(x)$.

Oscillator wave functions and the Hermite polynomials

At this level, we can show that the wave function (4.165) derived from the algebraic method is similar to the one obtained from the first method (4.118). To see this, we simply need to use the following operator identity:

$$e^{-x^2/2}\left(x-\frac{d}{dx}\right)e^{x^2/2}=-\frac{d}{dx}\quad\text{or}\quad e^{-x^2/2x_0^2}\left(x-x_0^2\frac{d}{dx}\right)e^{x^2/2x_0^2}=-x_0^2\frac{d}{dx}. \tag{4.166}$$

An application of this operator n times leads at once to

$$e^{-x^2/2x_0^2}\left(x-x_0^2\frac{d}{dx}\right)^n e^{x^2/2x_0^2}=(-1)^n(x_0^2)^n\frac{d^n}{dx^n}, \tag{4.167}$$

which can be shown to yield

$$\left(x-x_0^2\frac{d}{dx}\right)^n e^{-x^2/2x_0^2}=(-1)^n(x_0^2)^n e^{x^2/2x_0^2}\frac{d^n}{dx^n}e^{-x^2/x_0^2}. \tag{4.168}$$

We can now rewrite the right-hand side of this equation as follows:

$$\begin{aligned}(-1)^n(x_0^2)^n e^{x^2/2x_0^2}\frac{d^n}{dx^n}e^{-x^2/x_0^2} &= x_0^n e^{-x^2/2x_0^2}\left[(-1)^n e^{x^2/x_0^2}\frac{d^n}{d(x/x_0)^n}e^{-x^2/x_0^2}\right]\\ &= x_0^n e^{-x^2/2x_0^2}\left[(-1)^n e^{y^2}\frac{d^n}{dy^n}e^{-y^2}\right]\\ &= x_0^n e^{-x^2/2x_0^2}H_n(y),\end{aligned} \tag{4.169}$$

where $y=x/x_0$ and where $H_n(y)$ are the Hermite polynomials listed in (4.119):

$$H_n(y)=(-1)^n e^{y^2}\frac{d^n}{dy^n}e^{-y^2}. \tag{4.170}$$

Note that the polynomials $H_{2n}(y)$ are even and $H_{2n+1}(y)$ are odd, since $H_n(-y)=(-1)^n H_n(y)$.

Inserting (4.169) into (4.168), we obtain

$$\left(x-x_0^2\frac{d}{dx}\right)^n e^{-x^2/2x_0^2}=x_0^n e^{-x^2/2x_0^2}H_n\left(\frac{x}{x_0}\right); \tag{4.171}$$

substituting this equation into (4.165), we can write the oscillator wave function in terms of the Hermite polynomials as follows:

$$\boxed{\psi_n(x)=\frac{1}{\sqrt{\sqrt{\pi}2^n n!x_0}}e^{-x^2/2x_0^2}H_n\left(\frac{x}{x_0}\right).} \tag{4.172}$$

This wave function is identical with the one obtained from the first method (see Eq. (4.118)).

Remark

This wave function is either even or odd depending on n; in fact, the functions $\psi_{2n}(x)$ are even (i.e., $\psi_{2n}(-x)=\psi_{2n}(x)$) and $\psi_{2n+1}(x)$ are odd (i.e., $\psi_{2n}(-x)=-\psi_{2n}(x)$) since, as can be inferred from Eq (4.120), the Hermite polynomials $H_{2n}(x)$ are even and $H_{2n+1}(x)$ are odd. This is expected because, as mentioned in Section 4.2.4, the wave functions of even one-dimensional potentials have definite parity. Figure 4.9 displays the shapes of the first few wave functions.

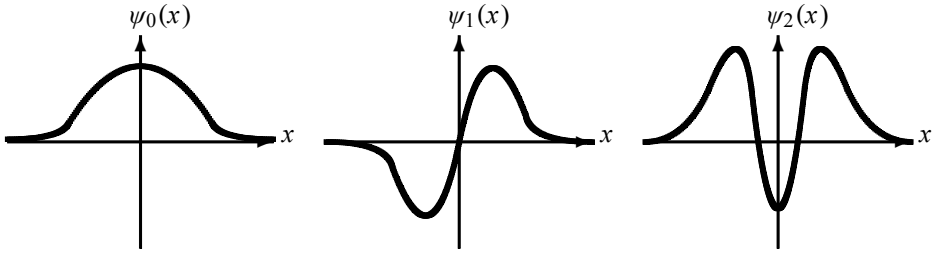


Figure 4.9 Shapes of the first three wave functions of the harmonic oscillator.

4.8.4 The Matrix Representation of Various Operators

Here we look at the matrix representation of several operators in the N-space. In particular, we focus on the representation of the operators $\hat{a}$, $\hat{a}^\dagger$, $\hat{X}$, and $\hat{P}$. First, since the states $|\, n\rangle$ are joint eigenstates of $\hat{H}$ and $\hat{N}$, it is easy to see from (4.130) and (4.132) that $\hat{H}$ and $\hat{N}$ are represented within the $\{|\, n\rangle\}$ basis by infinite diagonal matrices:

$$\langle n'|\hat{N} \mid n\rangle = n\delta_{n',n}, \qquad \langle n'|\hat{H} \mid n\rangle = \hbar\omega\left(n + \frac{1}{2}\right)\delta_{n',n}; \tag{4.173}$$

that is,

$$\hat{N} = \begin{pmatrix} 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & \cdots \\ 0 & 0 & 2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \qquad \hat{H} = \frac{\hbar\omega}{2}\begin{pmatrix} 1 & 0 & 0 & \cdots \\ 0 & 3 & 0 & \cdots \\ 0 & 0 & 5 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{4.174}$$

As for the operators $\hat{a}$, $\hat{a}^\dagger$, $\hat{X}$, $\hat{P}$, none of them are diagonal in the N-representation, since they do not commute with $\hat{N}$. The matrix elements of $\hat{a}$ and $\hat{a}^\dagger$ can be obtained from (4.143) and (4.144):

$$\langle n'|\hat{a} \mid n\rangle = \sqrt{n}\delta_{n',n-1}, \qquad \langle n'|\hat{a}^\dagger \mid n\rangle = \sqrt{n+1}\delta_{n',n+1}; \tag{4.175}$$

that is,

$$\hat{a} = \begin{pmatrix} 0 & \sqrt{1} & 0 & 0 & \cdots \\ 0 & 0 & \sqrt{2} & 0 & \cdots \\ 0 & 0 & 0 & \sqrt{3} & \cdots \\ 0 & 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \qquad \hat{a}^\dagger = \begin{pmatrix} 0 & 0 & 0 & 0 & \cdots \\ \sqrt{1} & 0 & 0 & 0 & \cdots \\ 0 & \sqrt{2} & 0 & 0 & \cdots \\ 0 & 0 & \sqrt{3} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{4.176}$$

Now, let us find the N-representation of the position and momentum operators, $\hat{X}$ and $\hat{P}$. From (4.122) we can show that $\hat{X}$ and $\hat{P}$ are given in terms of $\hat{a}$ and $\hat{a}^\dagger$ as follows:

$$\hat{X} = \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a} + \hat{a}^\dagger\right), \qquad \hat{P} = i\sqrt{\frac{m\hbar\omega}{2}}\left(\hat{a}^\dagger - \hat{a}\right). \tag{4.177}$$

Their matrix elements are given by

$$\langle n'|\hat{X} \mid n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\left(\sqrt{n}\delta_{n',n-1} + \sqrt{n+1}\delta_{n',n+1}\right), \tag{4.178}$$

$$\langle n'|\hat{P} \mid n\rangle = i\sqrt{\frac{m\hbar\omega}{2}}\left(-\sqrt{n}\delta_{n',n-1} + \sqrt{n+1}\delta_{n',n+1}\right), \tag{4.179}$$

in particular

$$\langle n \mid \hat{X} \mid n\rangle = \langle n \mid \hat{P} \mid n\rangle = 0. \tag{4.180}$$

The matrices corresponding to $\hat{X}$ and $\hat{P}$ are thus given by

$$\hat{X} = \sqrt{\frac{\hbar}{2m\omega}}\begin{pmatrix} 0 & \sqrt{1} & 0 & 0 & \cdots \\ \sqrt{1} & 0 & \sqrt{2} & 0 & \cdots \\ 0 & \sqrt{2} & 0 & \sqrt{3} & \cdots \\ 0 & 0 & \sqrt{3} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \tag{4.181}$$

$$\hat{P} = i\sqrt{\frac{m\hbar\omega}{2}}\begin{pmatrix} 0 & -\sqrt{1} & 0 & 0 & \cdots \\ \sqrt{1} & 0 & -\sqrt{2} & 0 & \cdots \\ 0 & \sqrt{2} & 0 & -\sqrt{3} & \cdots \\ 0 & 0 & \sqrt{3} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{4.182}$$

As mentioned in Chapter 2, the momentum operator is Hermitian, but not equal to its own complex conjugate: (4.182) shows that $\hat{P}^\dagger = \hat{P}$ and $\hat{P}^* = -\hat{P}$. As for $\hat{X}$, however, it is both Hermitian and equal to its complex conjugate: from (4.181) we have that $\hat{X}^\dagger = \hat{X}^* = \hat{X}$.

Finally, we should mention that the eigenstates $|n\rangle$ are represented by infinite column matrices; the first few states can be written as

$$|\,0\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ \vdots \end{pmatrix}, \qquad |\,1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ \vdots \end{pmatrix}, \qquad |\,2\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ \vdots \end{pmatrix}, \qquad |\,3\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ \vdots \end{pmatrix}, \ldots. \tag{4.183}$$

The set of states $\{|\,n\rangle\}$ forms indeed a complete and orthonormal basis.

4.8.5 Expectation Values of Various Operators

Let us evaluate the expectation values for $\hat{X}^2$ and $\hat{P}^2$ in the N-representation:

$$\hat{X}^2 = \frac{\hbar}{2m\omega}\left(\hat{a}^2 + \hat{a}^{\dagger 2} + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}\right) = \frac{\hbar}{2m\omega}\left(\hat{a}^2 + \hat{a}^{\dagger 2} + 2\hat{a}^\dagger\hat{a} + 1\right), \tag{4.184}$$

$$\hat{P}^2 = -\frac{m\hbar\omega}{2}\left(\hat{a}^2 + \hat{a}^{\dagger 2} - \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a}\right) = -\frac{m\hbar\omega}{2}\left(\hat{a}^2 + \hat{a}^{\dagger 2} - 2\hat{a}^\dagger\hat{a} - 1\right), \tag{4.185}$$

where we have used the fact that $\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} = 2\hat{a}^\dagger\hat{a} + 1$. Since the expectation values of $\hat{a}^2$ and $\hat{a}^{\dagger 2}$ are zero, $\langle n \mid \hat{a}^2 \mid n\rangle = \langle n \mid \hat{a}^{\dagger 2} \mid n\rangle = 0$, and $\langle n \mid \hat{a}^\dagger\hat{a} \mid n\rangle = n$, we have

$$\langle n \mid \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} \mid n\rangle = \langle n \mid 2\hat{a}^\dagger\hat{a} + 1 \mid n\rangle = 2n + 1; \tag{4.186}$$

hence

$$\langle n \mid \hat{X}^2 \mid n\rangle = \frac{\hbar}{2m\omega}\langle n \mid \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} \mid n\rangle = \frac{\hbar}{2m\omega}(2n+1), \tag{4.187}$$

$$\langle n \mid \hat{P}^2 \mid n\rangle = \frac{m\hbar\omega}{2}\langle n \mid \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} \mid n\rangle = \frac{m\hbar\omega}{2}(2n+1). \tag{4.188}$$

Comparing (4.187) and (4.188) we see that the expectation values of the potential and kinetic energies are equal and are also equal to half the total energy:

$$\boxed{\frac{m\omega^2}{2}\langle n \mid \hat{X}^2 \mid n\rangle = \frac{1}{2m}\langle n \mid \hat{P}^2 \mid n\rangle = \frac{1}{2}\langle n \mid \hat{H} \mid n\rangle.} \tag{4.189}$$

This result is known as the *Virial theorem*.

We can now easily calculate the product $\Delta x\,\Delta p$ from (4.187) and (4.188). Since $\langle\hat{X}\rangle = \langle\hat{P}\rangle = 0$ we have

$$\Delta x = \sqrt{\langle\hat{X}^2\rangle - \langle\hat{X}\rangle^2} = \sqrt{\langle\hat{X}^2\rangle} = \sqrt{\frac{\hbar}{2m\omega}(2n+1)}, \tag{4.190}$$

$$\Delta p = \sqrt{\langle\hat{P}^2\rangle - \langle\hat{P}\rangle^2} = \sqrt{\langle\hat{P}^2\rangle} = \sqrt{\frac{m\hbar\omega}{2}(2n+1)}; \tag{4.191}$$

hence

$$\Delta x\,\Delta p = \left(n + \frac{1}{2}\right)\hbar \quad\Longrightarrow\quad \Delta x\,\Delta p \geq \frac{\hbar}{2}, \tag{4.192}$$

since $n \geq 0$; this is the Heisenberg uncertainty principle.

4.9 Numerical Solution of the Schrödinger Equation

In this section we are going to show how to solve a one-dimensional Schrödinger equation numerically. The numerical solutions provide an idea about the properties of stationary states.

4.9.1 Numerical Procedure

We want to solve the following equation numerically:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi(x) = E\psi(x) \Longrightarrow \frac{d^2\psi}{dx^2} + k^2\psi(x) = 0, \tag{4.193}$$

where $k^2 = 2m[E - V(x)]/\hbar^2$.

First, divide the x-axis into a set of equidistant points with a spacing of $h_0 = \Delta x$, as shown in Figure 4.10a. The wave function $\psi(x)$ can be approximately described by its values at the

Chapter 6

Three-Dimensional Problems

6.1 Introduction

In this chapter we examine how to solve the Schrödinger equation for spinless particles moving in three-dimensional potentials. We carry out this study in two different coordinate systems: the Cartesian system and the spherical system.

First, working within the context of Cartesian coordinates, we study the motion of a particle in different potentials: the free particle, a particle in a (three-dimensional) rectangular potential, and a particle in a harmonic oscillator potential. This study is going to be a simple generalization of the one-dimensional problems presented in Chapter 4. Unlike the one-dimensional case, three-dimensional problems often exhibit degeneracy, which occurs whenever the potential displays symmetry.

Second, using spherical coordinates, we describe the motion of a particle in spherically symmetric potentials. After presenting a general treatment, we consider several applications ranging from the free particle and the isotropic harmonic oscillator to the hydrogen atom. We conclude the chapter by calculating the energy levels of a hydrogen atom when placed in a constant magnetic field; this gives rise to the Zeeman effect.

6.2 3D Problems in Cartesian Coordinates

We examine here how to extend Schrödinger's theory of one-dimensional problems (Chapter 4) to three dimensions.

6.2.1 General Treatment: Separation of Variables

The time-dependent Schrödinger equation for a spinless particle of mass m moving under the influence of a three-dimensional potential is

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\Psi(x,y,z,t)+\hat{V}(x,y,z,t)\Psi(x,y,z)=i\hbar\frac{\partial\Psi(x,y,z,t)}{\partial t}, \tag{6.1}$$

where $\vec{\nabla}^2$ is the Laplacian, $\vec{\nabla}^2=\partial^2/\partial x^2+\partial^2/\partial y^2+\partial^2/\partial z^2$. As seen in Chapter 4, the wave function of a particle moving in a time-independent potential can be written as a product of

spatial and time components:

$$\Psi(x, y, z, t) = \psi(x, y, z)e^{-iEt/\hbar}, \tag{6.2}$$

where $\psi(x, y, z)$ is the solution to the time-independent Schrödinger equation:

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(x, y, z) + \hat{V}(x, y, z)\psi(x, y, z) = E\psi(x, y, z), \tag{6.3}$$

which is of the form $\hat{H}\psi = E\psi$.

This partial differential equation is generally difficult to solve. But, for those cases where the potential $\hat{V}(x, y, z)$ separates into the sum of three independent, one-dimensional terms (which should not be confused with a vector)

$$V(x, y, z) = V_x(x) + V_y(y) + V_z(z), \tag{6.4}$$

we can solve (6.3) by means of the technique of *separation of variables*. This technique consists of separating the three-dimensional Schrödinger equation (6.3) into three independent one-dimensional Schrödinger equations. Let us examine how to achieve this. Note that (6.3), in conjunction with (6.4), can be written as

$$\left[\hat{H}_x + \hat{H}_y + \hat{H}_z\right]\psi(x, y, z) = E\psi(x, y, z), \tag{6.5}$$

where $\hat{H}_x$ is given by

$$\hat{H}_x = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V_x(x); \tag{6.6}$$

the expressions for $\hat{H}_y$ and $\hat{H}_z$ are analogous.

As $\hat{V}(x, y, z)$ separates into three independent terms, we can also write $\psi(x, y, z)$ as a product of three functions of a single variable each:

$$\psi(x, y, z) = X(x)Y(y)Z(z). \tag{6.7}$$

Substituting (6.7) into (6.5) and dividing by $X(x)Y(y)Z(z)$, we obtain

$$\left[-\frac{\hbar^2}{2m}\frac{1}{X}\frac{d^2X}{dx^2} + V_x(x)\right] + \left[-\frac{\hbar^2}{2m}\frac{1}{Y}\frac{d^2Y}{dy^2} + V_y(y)\right] + \left[-\frac{\hbar^2}{2m}\frac{1}{Z}\frac{d^2Z}{dz^2} + V_z(z)\right] = E. \tag{6.8}$$

Since each expression in the square brackets depends on only one of the variables x, y, z, and since the sum of these three expressions is equal to a constant, E, each separate expression must then be equal to a constant such that the sum of these three constants is equal to E. For instance, the x-dependent expression is given by

$$\left[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V_x(x)\right]X(x) = E_xX(x). \tag{6.9}$$

Similar equations hold for the y and z coordinates, with

$$E_x + E_y + E_z = E. \tag{6.10}$$

The separation of variables technique consists in essence of reducing the three-dimensional Schrödinger equation (6.3) into three separate one-dimensional equations (6.9).

6.2.2 The Free Particle

In the simple case of a free particle, the Schrödinger equation (6.3) reduces to three equations similar to (6.9) with $V_x = 0$, $V_y = 0$, and $V_z = 0$. The x-equation can be obtained from (6.9):

$$\frac{d^2X(x)}{dx^2} = -k_x^2 X(x), \tag{6.11}$$

where $k_x^2 = 2mE_x/\hbar^2$, and hence $E_x = \hbar^2 k_x^2/(2m)$. As shown in Chapter 4, the normalized solutions to (6.11) are plane waves

$$X(x) = \frac{1}{\sqrt{2\pi}} e^{ik_x x}. \tag{6.12}$$

Thus, the solution to the three-dimensional Schrödinger equation (6.3) is given by

$$\psi_{\vec{k}}(x, y, z) = (2\pi)^{-3/2} e^{ik_x x} e^{ik_y y} e^{ik_z z} = (2\pi)^{-3/2} e^{i\vec{k}\cdot\vec{r}}, \tag{6.13}$$

where $\vec{k}$ and $\vec{r}$ are the wave and position vectors of the particle, respectively. As for the total energy E, it is equal to the sum of the eigenvalues of the three one-dimensional equations (6.11):

$$E = E_x + E_y + E_z = \frac{\hbar^2}{2m}\left(k_x^2 + k_y^2 + k_z^2\right) = \frac{\hbar^2}{2m}\vec{k}^2. \tag{6.14}$$

Note that, since the energy (6.14) depends only on the magnitude of $\vec{k}$, all different orientations of $\vec{k}$ (obtained by varying k_x, k_y, k_z) subject to the condition

$$|\vec{k}| = \sqrt{k_x^2 + k_y^2 + k_z^2} = constant \tag{6.15}$$

generate different eigenfunctions (6.13) without a change in the energy. As the total number of orientations of $\vec{k}$ which preserve its magnitude is infinite, the energy of a free particle is *infinitely degenerate*.

Note that the solutions to the time-dependent Schrödinger equation (6.1) are obtained by substituting (6.13) into (6.2):

$$\Psi_{\vec{k}}(\vec{r}, t) = \psi(\vec{r}) e^{-i\omega t} = (2\pi)^{-3/2} e^{i(\vec{k}\cdot\vec{r} - \omega t)}, \tag{6.16}$$

where $\omega = E/\hbar$; this represents a propagating wave with wave vector $\vec{k}$. The orthonormality condition of this wave function is expressed by

$$\int \Psi^*_{\vec{k}'}(\vec{r}, t)\Psi_{\vec{k}}(\vec{r}, t)\, d^3r = \int \psi^*_{\vec{k}'}(\vec{r})\psi_{\vec{k}}(\vec{r})\, d^3r = (2\pi)^{-3} \int e^{i(\vec{k}-\vec{k}')\cdot\vec{r}} d^3r = \delta(\vec{k} - \vec{k}'), \tag{6.17}$$

which can be written in Dirac's notation as

$$\langle \Psi_{\vec{k}'}(t) | \Psi_{\vec{k}}(t) \rangle = \langle \psi_{\vec{k}'} | \psi_{\vec{k}} \rangle = \delta(\vec{k} - \vec{k}'). \tag{6.18}$$

The free particle can be represented, as seen in Chapter 3, by a wave packet (a superposition of wave functions corresponding to the various wave vectors):

$$\Psi(\vec{r}, t) = (2\pi)^{-3/2} \int A(\vec{k}, t)\Psi_{\vec{k}}(\vec{r}, t)\, d^3k = (2\pi)^{-3/2} \int A(\vec{k}, t) e^{i(\vec{k}\cdot\vec{r} - \omega t)} d^3k, \tag{6.19}$$

where $A(\vec{k}, t)$ is the Fourier transform of $\Psi(\vec{r}, t)$:

$$A(\vec{k}, t) = (2\pi)^{-3/2} \int \Psi(\vec{r}, t) e^{-i(\vec{k}\cdot\vec{r} - \omega t)} d^3r. \tag{6.20}$$

As seen in Chapters 1 and 4, the position of the particle can be represented classically by the center of the wave packet.

6.2.3 The Box Potential

We are going to begin with the rectangular box potential, which has no symmetry, and then consider the cubic potential, which displays a great deal of symmetry, since the xyz axes are equivalent.

6.2.3.1 The Rectangular Box Potential

Consider first the case of a spinless particle of mass m confined in a *rectangular* box of sides $a,\ b,\ c$:

$$V(x, y, z) = \begin{cases} 0, & 0 < x < a,\ 0 < y < b,\ 0 < z < c, \\ \infty, & \text{elsewhere,} \end{cases} \tag{6.21}$$

which can be written as $V(x, y, z) = V_x(x) + V_y(y) + V_z(z)$, with

$$V_x(x) = \begin{cases} 0, & 0 < x < a, \\ \infty, & \text{elsewhere;} \end{cases} \tag{6.22}$$

the potentials $V_y(y)$ and $V_z(z)$ have similar forms.

The wave function $\psi(x, y, z)$ must vanish at the walls of the box. We have seen in Chapter 4 that the solutions for this potential are of the form

$$X(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n_x \pi}{a} x\right), \qquad n_x = 1, 2, 3, \ldots, \tag{6.23}$$

and the corresponding energy eigenvalues are

$$E_{n_x} = \frac{\hbar^2 \pi^2}{2ma^2} n_x^2. \tag{6.24}$$

From these expressions we can write the normalized three-dimensional eigenfunctions and their corresponding energies:

$$\boxed{\psi_{n_x n_y n_z}(x, y, z) = \sqrt{\frac{8}{abc}} \sin\left(\frac{n_x \pi}{a} x\right) \sin\left(\frac{n_y \pi}{b} y\right) \sin\left(\frac{n_z \pi}{c} z\right),} \tag{6.25}$$

$$\boxed{E_{n_x n_y n_z} = \frac{\hbar^2 \pi^2}{2m} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2}\right).} \tag{6.26}$$

Table 6.1 Energy levels and their degeneracies for the cubic potential, with $E_1 = \frac{\pi^2\hbar^2}{2mL^2}$.

$E_{n_x n_y n_z}/E_1$	$(n_x,\ n_y,\ n_z)$	g_n
3	(111)	1
6	(211), (121), (112)	3
9	(221), (212), (122)	3
11	(311), (131), (113)	3
12	(222)	1
14	(321), (312), (231), (213), (132), (123)	6

6.2.3.2 The Cubic Potential

For the simpler case of a *cubic* box of side L, the energy expression can be inferred from (6.26) by substituting $a = b = c = L$:

$$\boxed{E_{n_x n_y n_z} = \frac{\hbar^2\pi^2}{2mL^2}(n_x^2 + n_y^2 + n_z^2), \qquad n_x,\ n_y,\ n_z = 1, 2, 3, \ldots.} \tag{6.27}$$

The ground state corresponds to $n_x = n_y = n_z = 1$; its energy is given by

$$E_{111} = \frac{3\pi^2\hbar^2}{2mL^2} = 3E_1, \tag{6.28}$$

where, as shown in Chapter 4, $E_1 = \pi^2\hbar^2/(2mL^2)$ is the zero-point energy of a particle in a one-dimensional box. Thus, the zero-point energy for a particle in a three-dimensional box is three times that in a one-dimensional box. The factor 3 can be viewed as originating from the fact that we are confining the particle symmetrically in all three dimensions.

The first excited state has three possible sets of quantum numbers $(n_x,\ n_y,\ n_z) = (2, 1, 1)$, $(1, 2, 1)$, $(1, 1, 2)$ corresponding to three different states $\psi_{211}(x, y, z)$, $\psi_{121}(x, y, z)$, and $\psi_{112}(x, y, z)$, where

$$\psi_{211}(x, y, z) = \sqrt{\frac{8}{L^3}} \sin\left(\frac{2\pi}{L}x\right) \sin\left(\frac{\pi}{L}y\right) \sin\left(\frac{\pi}{L}z\right); \tag{6.29}$$

the expressions for $\psi_{121}(x, y, z)$ and $\psi_{112}(x, y, z)$ can be inferred from $\psi_{211}(x, y, z)$. Notice that all three states have the same energy:

$$E_{211} = E_{121} = E_{112} = 6\frac{\pi^2\hbar^2}{2mL^2} = 6E_1. \tag{6.30}$$

The first excited state is thus threefold *degenerate*.

Degeneracy occurs only when there is a symmetry in the problem. For the present case of a particle in a cubic box, there is a great deal of symmetry, since all three dimensions are equivalent. Note that for the rectangular box, there is no degeneracy since the three dimensions are not equivalent. Moreover, degeneracy did not exist when we treated one-dimensional problems in Chapter 4, for they give rise to only one quantum number.

The second excited state also has three different states, and hence it is threefold degenerate; its energy is equal to $9E_1$: $E_{221} = E_{212} = E_{122} = 9E_1$.

The energy spectrum is shown in Table 6.1, where every nth level is characterized by its energy, its quantum numbers, and its degeneracy g_n.

6.2.4 The Harmonic Oscillator

We are going to begin with the anisotropic oscillator, which displays no symmetry, and then consider the isotropic oscillator where the xyz axes are all equivalent.

6.2.4.1 The Anisotropic Oscillator

Consider a particle of mass m moving in a three-dimensional anisotropic oscillator potential

$$\hat{V}(\hat{x},\hat{y},\hat{z}) = \frac{1}{2}m\omega_x^2\hat{X}^2 + \frac{1}{2}m\omega_y^2\hat{Y}^2 + \frac{1}{2}m\omega_z^2\hat{Z}^2. \tag{6.31}$$

Its Schrödinger equation separates into three equations similar to (6.9):

$$-\frac{\hbar^2}{2m}\frac{d^2X(x)}{dx^2} + \frac{1}{2}m\omega_x x^2 X(x) = E_x X(x), \tag{6.32}$$

with similar equations for $Y(y)$ and $Z(z)$. The eigenenergies corresponding to the potential (6.31) can be expressed as

$$\boxed{E_{n_xn_yn_z} = E_{n_x} + E_{n_y} + E_{n_z} = \left(n_x + \frac{1}{2}\right)\hbar\omega_x + \left(n_y + \frac{1}{2}\right)\hbar\omega_y + \left(n_z + \frac{1}{2}\right)\hbar\omega_z,} \tag{6.33}$$

with $n_x, n_y, n_z = 0, 1, 2, 3, \ldots$. The corresponding stationary states are

$$\psi_{n_xn_yn_z}(x,y,z) = X_{n_x}(x)Y_{n_y}(y)Z_{n_z}(z), \tag{6.34}$$

where $X_{n_x}(x)$, $Y_{n_y}(y)$, and $Z_{n_z}(z)$ are one-dimensional harmonic oscillator wave functions. These states are not degenerate, because the potential (6.31) has no symmetry (it is anisotropic).

6.2.4.2 The Isotropic Harmonic Oscillator

Consider now an *isotropic* harmonic oscillator potential. Its energy eigenvalues can be inferred from (6.33) by substituting $\omega_x = \omega_y = \omega_z = \omega$,

$$\boxed{E_{n_xn_yn_z} = \left(n_x + n_y + n_z + \frac{3}{2}\right)\hbar\omega.} \tag{6.35}$$

Since the energy depends on the sum of n_x, n_y, n_z, any set of quantum numbers having the same sum will represent states of equal energy.

The ground state, whose energy is $E_{000} = 3\hbar\omega/2$, is not degenerate. The first excited state is threefold degenerate, since there are three different states, ψ_{100}, ψ_{010}, ψ_{001}, that correspond to the same energy $5\hbar\omega/2$. The second excited state is sixfold degenerate; its energy is $7\hbar\omega/2$.

In general, we can show that the degeneracy g_n of the nth excited state, which is equal to the number of ways the nonnegative integers n_x, n_y, n_z may be chosen to total to n, is given by

$$\boxed{g_n = \frac{1}{2}(n+1)(n+2),} \tag{6.36}$$

where $n = n_x + n_y + n_z$. Table 6.2 displays the first few energy levels along with their degeneracies.

Table 6.2 Energy levels and their degeneracies for an isotropic harmonic oscillator.

n	$2E_n/(\hbar\omega)$	$(n_x n_y n_z)$	g_n
0	3	(000)	1
1	5	(100), (010), (001)	3
2	7	(200), (020), (002)	6
		(110), (101), (011)	
3	9	(300), (030), (003)	10
		(210), (201), (021)	
		(120), (102), (012)	
		(111)	

Example 6.1 (Degeneracy of a harmonic oscillator)
Show how to derive the degeneracy relation (6.36).

Solution
For a fixed value of n, the degeneracy g_n is given by the number of ways of choosing n_x, n_y, and n_z so that $n = n_x + n_y + n_z$.

For a fixed value of n_x, the number of ways of choosing n_y and n_z so that $n_y + n_z = n - n_x$ is given by $(n - n_x + 1)$; this can be shown as follows. For a given value of n_x, the various permissible values of (n_y, n_z) are given by $(n_y, n_z) = (0, n-n_x)$, $(1, n-n_x-1)$, $(2, n-n_x-2)$, $(3, n - n_x - 3)$, …, $(n - n_x - 3, 3)$, $(n - n_x - 2, 2)$, $(n - n_x - 1, 1)$, and $(n - n_x, 0)$. In all, there are $(n - n_x + 1)$ sets of (n_y, n_z) so that $n_y + n_z = n - n_x$. Now, since the values of n_x can vary from 0 to n, the degeneracy is then given by

$$g_n = \sum_{n_x=0}^{n} (n - n_x + 1) = (n+1) \sum_{n_x=0}^{n} 1 - \sum_{n_x=0}^{n} n_x = (n+1)^2 - \frac{1}{2} n(n+1) = \frac{1}{2}(n+1)(n+2). \tag{6.37}$$

A more primitive way of calculating this series is to use Gauss's method: simply write the series $\sum_{n_x=0}^{n}(n - n_x + 1)$ in the following two equivalent forms:

$$g_n = (n+1) + n + (n-1) + (n-2) + \cdots + 4 + 3 + 2 + 1, \tag{6.38}$$

$$g_n = 1 + 2 + 3 + 4 + \cdots + (n-2) + (n-1) + n + (n+1). \tag{6.39}$$

Since both of these two series contain $(n + 1)$ terms each, a term by term addition of these relations yields

$$\begin{aligned} 2g_n &= (n+2) + (n+2) + (n+2) + \cdots + (n+2) + (n+2) + (n+2) \\ &= (n+1)(n+2); \end{aligned} \tag{6.40}$$

hence $g_n = \frac{1}{2}(n + 1)(n + 2)$.